

Volume 50

Sentinel Data Protection & Privacy Engine (SDPPE)

Status: Core Security Domain

Vision

Project Sentinel must protect not only systems, but also the **data inside and around those systems**.

The platform should answer:

- What sensitive data exists?
 - Where is it stored?
 - Who can access it?
 - Is it encrypted?
 - Is it backed up?
 - Is it exposed?
 - Is it governed by policy?
 - Is privacy respected?
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Data Philosophy

Data is not just files.

Data includes:

Documents

Every data source becomes part of the Digital Twin where appropriate.

Data Object

Every important data collection can become a Sentinel Object.

Data Object

Data Classification

Initial classifications:

Public

Organizations can define custom classifications.

Privacy by Design

Project Sentinel must minimize unnecessary collection.

The platform should collect:

- only what is needed,

- only with authorization,
 - only for a defined purpose,
 - only for the required retention period.
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Sensitive Data Awareness

The engine can help identify categories such as:

- Personal information
- Credentials
- API keys
- Financial information
- Health information
- Legal documents
- Customer data
- Security evidence
- Internal reports

Detection must be careful, configurable, and evidence-based.

Encryption Verification

The engine checks:

- Disk encryption
 - Database encryption
 - Backup encryption
 - Cloud storage encryption
 - Transport encryption
 - Key management status
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Access Intelligence

The platform answers:

Who can access this data?

Data Retention

Organizations define retention policies.

Examples:

- Evidence retained for 12 months.
 - Incident data retained for 7 years.
 - Personal data minimized.
 - Reports archived after review.
 - Logs rotated according to policy.
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Data Privacy Dashboard

For home users:

Your data protection

For enterprises:

Data Coverage

NEW CORE ENGINE

Data Classification Engine

Classifies data objects based on metadata, source, owner, policy, and evidence.

NEW CORE ENGINE

Data Exposure Engine

Detects authorized evidence of exposure such as:

- Public storage
 - Overly broad permissions
 - Unencrypted transfer
 - Weak access policies
 - Unreviewed sharing links
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NEW CORE ENGINE

Privacy Governance Engine

Maps privacy requirements to:

- data objects,
 - systems,
 - users,
 - policies,
 - evidence,
 - retention periods.
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NEW CORE ENGINE

Backup Verification Engine

Checks whether critical data has recoverable backup evidence.

It should verify backup status where integrations allow it, without assuming recovery unless tested.

PS-ADR-0045

Data Protection Is a First-Class Security Domain

Status: Accepted

Decision

Project Sentinel shall treat data, privacy, retention, encryption, backup, access, and exposure as first-class parts of the Cyber Security Operating Environment.

Rationale

Security cannot be measured only by device or network posture. The true objective is protecting people, organizations, operations, and data.

Strategic Principle

Project Sentinel protects systems so that people and data remain protected.

Next volume: **Volume 51 - Frontend & User Experience Architecture**, where we define how the whole system stays simple, accessible, multilingual, and easy to use.

====[USER]====

good and continue

====[ASSISTANT]====

Excellent.

PROJECT SENTINEL BLUEPRINT

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